

SCIENCE (Code 086) — Sample Question Paper Excerpt

CLASS X (2025-26) — Test subset: Q10, Q13, Q28

(Extracted from the official CBSE Class X Science SQP 2025-26, for pipeline testing)

- 10.** Unlike animals, plants do not have any excretory products as they do not eat food. Comment upon the statement with justification. **2**
- 13.** Draw and explain how the nerve cells help in transmission of impulses? **3**
- 28.** Sara took 2 mL of dilute NaOH solution in a test tube and added two drops of phenolphthalein solution to it. The solution turned pink in colour. She added dilute H_2SO_4 to the above solution drop by drop until the solution in the test tube became colourless. 40 drops of dilute H_2SO_4 were used for the change in colour from pink to colourless. When Sara added a drop of NaOH to the solution, the colour changed back to pink again. **4**

Sara now tried the activity with different volumes of NaOH and recorded her observations in the table below:

S. No.	Volume of dil. NaOH taken (mL)	Drops of dil. H_2SO_4 used
1	2	20
2	3	30
3	4	40

Answer the following questions based on the above information:

A. If Sara used concentrated H_2SO_4 in place of dilute H_2SO_4 , how many drops will be required for the change in colour to be observed? (a) 40 (b) < 40 (c) > 40 — Justify your answer.

B. Sara measured 20 drops of dil. H_2SO_4 and found its volume to be 1 mL. If Sara observed a change in colour of NaOH solution by using 3 mL of H_2SO_4 , how many mL of NaOH did she add to the test tube initially?

C. Write a balanced chemical equation for the reaction taking place in the above experiment. Which of the following is true and why? The reaction is a (a) neutralisation and double displacement reaction (b) neutralisation and precipitation reaction (c) precipitation and double displacement reaction (d) neutralisation, double displacement as well as precipitation reaction.